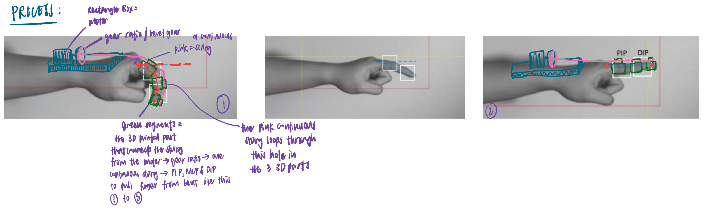
Biomechanics of a single finger

→ Extensor : located on the back of hand → extends index finger

→ flexor : located on the palm side → flexes at/z of index finger

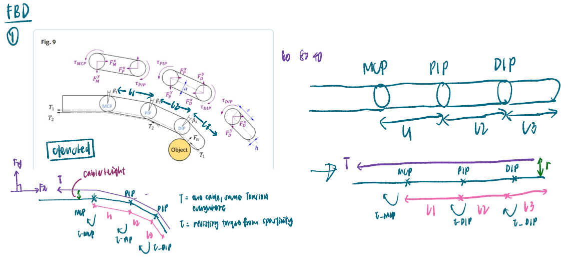
PROCESS :



Green segments = the 3D printed part that connects the string from the motor → gear ratio → one continuous string → PIP, MCP & DIP to pull finger from bent like this ① to ②

the pink continuous string loops through this hole in the 3 3D parts

FBD



T = one cable, same tension everywhere

τ = resisting torque from spasticity

✱ Is the MG996R strong enough to open a stroke patient's (S2–S3) hand?

→ A servo motor on the forearm winds the fishing line onto a spool. The fishing line runs over the back of the hand of each finger through 3D printed PLA guides & straightens the clenched fingers (① → ②) FBD:

→ T : pull (tension) the line needs to open the fingers

Friction : How much extra pull the PLA guides add

Cable Travel : How much line must be reeled in

Motor Torque : pull × spool radius (compare w/ what the MG996R can give)

1. Key Ideas

	units :
– Tension (T) : pull carried along a string (same along one string if friction is ignored)	N
– Torque (τ) : turning effect = force × ⊥ dist from pivot (spanner)	N·m
– Moment Arm (r) : ⊥ dist from joint centre to cable	m
– Friction Factor : Extra pull added when cable rubs on PLA guides ($e^{\mu\beta}$) ↓ capstan equ	–
– Spool Radius : radius of the drum the line winds on (Motor Torque = pull × spool radius)	m
– Stall torque : most torque the servo gives before it stops	N·m
– Safety Factor (FoS) : Multiplier so device works for patients stronger than those measured	–

2. From the FBD

→ the purple line w/ T is the one tension pull towards motor

→ At every joint, cable's turning effect T×r opens the finger

→ τ patient's spasticity resist

→ Joint opens when T×r ≥ τ

3. Where the force comes from

- Alhamed, Seth & Abdullah (2023) built a cable-driven hand demo in which one motor pulls the 4 fingers (index → pinky) as a group through a cable w/ a load sensor measuring the cable force. W/ 8 stroke (stage 2–3) patients, they measured the force needed to extend the impaired fingers.
- the paper states that these forces are the tension in the cable used to open the hand
- Average : 15.07N

  Max : 25.48N

4. Values

- Tavg = 15.07 N ; Tmax = 25.48N } cable Tension
- Safety factor used in the paper: FoS = 2
- l1 ; l2 ; l3 = 26.7 ; 22.4 ; 26.4 mm
- Finger thickness (h) = 16.2 ; 14.0 ; 13.6 mm
- Guide hole = 9.7mm (Hole diameter where cable loops through)

- clenched angles : θmcp = 80° θpip = 60° θdip = 40°
- μ(fishing line on PLA) = 0.3 } Assumed ; PLA sliding friction measured (Sirin et al ; 2024)
- MG996R stall torque : 9.4kgf·cm (4.8V)

  11 kgf·cm (6V)
- MG996R rotation : 180° [standard positional servo]

4. Workings

S1: cable tension to open 4 fingers

Tavg = 15.07N Tmax = 25.48N

$T_{finger\,avg} = \frac{15.07}{4} = 3.77N$ (3sf)

$T_{finger\,max} = \frac{25.48N}{4} = 6.37N$ (3sf)

S2: cable leverage at each joint when clenched (FBD1)

→ shows how our guide geometry gives the cable leverage to open at each joint, & where the cable runs.

→ straight like FBD 2 : r = 9.7mm + h/2 ⇒ $9.7 + \frac{16.2}{2} = 17.8$; $9.7 + \frac{14}{2} = 16.7$; $9.7 + \frac{13.6}{2} = 16.5$

↳ bent like FBD 1 : r = [(a₁d₂ + a₂d₁) cosθ + (d₁d₂ − a₁a₂) sinθ] ÷ chord

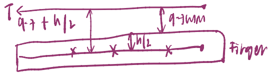
↳ cable runs straight btwn guides

↳ d = guide height above joint axis

a = guide distance from joint [segment midpoints]

θ = joint angle

chord = distance btwn 2 guide holes



rmcp

since there's no guide before it,

$r = \frac{h}{2} = \frac{16.2}{2} = 8.1mm$

rpip

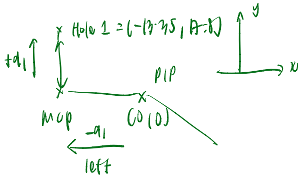
$a_1 = \frac{l1}{2} = \frac{26.7}{2} = 13.35mm$ $d_1 = 9.7 + \frac{h_1}{2} = 17.8$ mm

$a_2 = \frac{l2}{2} = \frac{22.4}{2} = 11.2mm$ $d_2 = 9.7 + \frac{h_2}{2} = 16.7$ m

Hole 1 = (−a₁, d₁) = (−13.35, 17.8)

Hole 2 = (a₂cosθ + d₂sinθ , −a₂sinθ + d₂cosθ)

= (11.2 × cos(60) + 16.7 × sin(60), −11.2 × sin(60) + 16.7 × cos(60)) = (20.06, −1.35)



chord

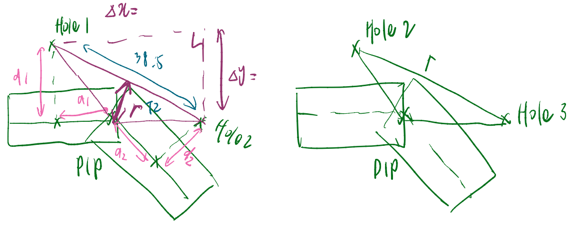
Δx = 20.06 − (−13.35) = 33.41

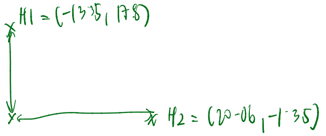
Δy = 17.80 − (−1.35) = 19.15

chord = $\sqrt{(33.41)^2 + (19.15)^2}$ = 38.5mm

½ × b × h = area

½ × 38.5 × r = area → r = area × 2 ÷ 38.5





area of Δ btwn 3 points = ½ |x₁y₂ − x₂y₁|

= ½ |(13.35 × 1.35) − (20.06 × 17.8)|

= 169.52275

→ $r = \frac{(169.52275) \times 2}{38.5}$ = 8.8mm

S3: friction in our PLA guides

→ our line bends through PLA guides & over knuckle (paper device did not)

→ motor must pull extra to overcome friction

} μ ≠ 0 (coz friction exists & Hence not ideal)

$F_{motor} = T_{fingers} \times e^{\mu\beta}$ (4 fingers)

μ = friction coefficient

β = total bend of lines in radians

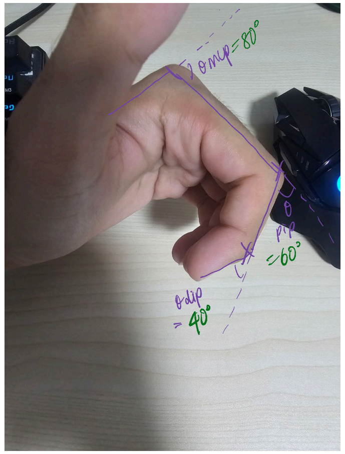
= 80° + 60° + 40° = 180°

= 3.142 radians

μ = 0.3 (realistic)

$F_{motor} = (15.07)(e^{0.3 \times 3.142}) = 38.7N$ (avg)

$F_{motor\,max} = (25.4)(e^{0.3 \times 3.142}) = 65.4N$



S5 : spool radius

→ MG996R only rotates 180° (π rad)

s = rs × θservo → rs = s ÷ π

= 35.1mm ÷ 3.142

= 11.2mm (minimum)

→ smaller than 11.2mm : servo runs out of rotation b4 hand is fully open

→ larger than 11.2mm : open hands fully but servo needs more torque

S6: Motor torque needed:

→ line pulls on the edge of the spool. servo must overcome that turning effect

→ τmotor = Fmotor × rs (step 3) → rs = 0.0112m

τAvg = (38.7)(0.0112m) = 0.43N·m τMax = (65.4)(0.0112) = 0.73N·m

S7: What MG996R can give

→ convert to N·m

→ stall : absolute limit → half taken as safe working torque

τ(N·m) = kgf·cm × 9.81 × 0.01

9.81 N per kgf ; 0.01m per cm

6V : 11 × 9.81 × 0.01 = 1.08N·m

4.8V : 9.4 × 9.81 × 0.01 = 0.92N·m

1.08 N·m (stall torque)

safety = $\frac{1.08}{2}$ = 0.54N·m

safety = $\frac{0.92}{2}$ = 0.46N·m

S8: Compare

Case	Motor Pull	Torque Needed	6V (Safe 0.54 / Stall 1.08)	4.8V (Safe 0.46 / Stall 0.92)
Average patient (15.07N)	38.7N	0.43N·m	✓ sufficient	✓ sufficient (just nice)
Worst Patient (25.48N)	65.4N	0.73N·m	– over safe, under stall	– over safe; under stall
Avg pt × safety factor 2	77.3N	0.87N·m	– over safe, under stall	– over safe; under stall
worst pt × safety factor 2	130.8N	1.46 N·m	✗ exceeds stall	✗ exceeds stall

✱ prev : Assumes if MG996R rotates 180° (fully)

✱ if min 160° rotation w/ μ = 0.3

Case	Motor Pull	Torque Needed	6V (Safe 0.54 / Stall 1.08)	4.8V (Safe 0.46 / Stall 0.92)
Average patient (15.07N)	38.7N	38.7 × 0.0126 = 0.49	✓ suff; below safe	over safe ; under stall
Worst Patient (25.48N)	65.4N	65.4 × 0.0126 = 0.82	over safe ; under stall	over safe ; under stall
Avg pt × safety factor 2	77.3N	77.3 × 0.0126 = 0.97	over safe ; under stall	✗
worst pt × safety factor 2	130.8N	130.8 × 0.0126 = 1.65	✗	✗

step 4: cable travel

→ the spool must reel in all the line the finger releases when it straightens

→ sets spool size

MCP : s = rmcp × θ (line wraps over knuckle) — θ in radians

= 8.1 × 1.396 = 11.3

PIP ; DIP ⇒ s = chord(bent) − chord(straight)

Spip = 38.5 − 24.6 = 13.9

Sdip = 34.3 − 24.4 = 9.9

Stotal = 13.9 + 9.9 + 11.3

= 35.1mm

rdip

Hole 2 = (11.2, 16.7)

Hole 3 = (13.2 × cos40 + 16.5 × sin40, −13.2 × sin40 + 16.5 × cos40)

= (20.73, 4.15)

chord = $\sqrt{(31.93)^2 + (12.55)^2}$ = 34.3mm

2 × area = |(−11.2)(4.15) − (20.73)(16.7)| = 392.5

$r\_dip = \frac{392.5}{34.3}$ = 11.4mm